

PARTIAL DERIVATIVES

- Let $z = f(x, y)$ be a function of two variables in a domain $D \subset R$. Let $(a, b) \in D$
 - (1) If $\lim_{h \rightarrow 0} \frac{f(a+h, b) - f(a, b)}{h}$ exists, then it is said to be the partial derivative of $f(x, y)$ with respect to x at the point (a, b) . It is denoted by $\left[\frac{\partial f}{\partial x} \right]_{(a, b)}$ or $\left[\frac{\partial z}{\partial x} \right]_{(a, b)}$ or $f_x(a, b)$.
 - (2) If $\lim_{h \rightarrow 0} \frac{f(a, b+h) - f(a, b)}{h}$ exists, then it is said to be the partial derivative of $f(x, y)$ with respect to y at the point (a, b) . It is denoted by $\left[\frac{\partial f}{\partial y} \right]_{(a, b)}$ or $\left[\frac{\partial z}{\partial y} \right]_{(a, b)}$ or $f_y(a, b)$.
 - (3) Usually the partial derivative of $z = f(x, y)$ at a point (x, y) are denoted by $\frac{\partial f}{\partial x}$ or f_x and $\frac{\partial f}{\partial y}$ or f_y .
 - (4) Partial derivative of $z = f(x, y)$ with respect to 'x' is the ordinary derivative of $z = f(x, y)$ with respect to 'x' by taking 'y' as a constant.
 - (5) Partial derivative of $z = f(x, y)$ with respect to 'y' is the ordinary derivative of $z = f(x, y)$ with respect to 'y' by taking 'x' as a constant.
 - (6) If $z = f(x, y)$ then the first order partial derivatives are $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$.
 - (7) If $z = f(x, y)$ then the second order partial derivatives are $\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = f_{xx}$,
 $\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right) = f_{yy}$, $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) = f_{xy}$ and $\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right) = f_{yx}$.
- If $u = f(z)$ and $z = f(x, y)$ then $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial z} \frac{\partial z}{\partial x}$ and $\frac{\partial u}{\partial y} = \frac{\partial u}{\partial z} \frac{\partial z}{\partial y}$.
- If $z = f(x, y)$ then $dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$.
- If $z = f(x, y)$ and $x = g(t), y = h(t)$ then $\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$.
- If $z = f(u, v)$ and $u = g(x, y), v = h(x, y)$ then $\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x}$, $\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y}$.
- If $f(x, y) = c$ is an implicit function, then $\frac{dy}{dx} = - \frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} = - \frac{f_x}{f_y}$.
- **Homogeneous Function of Degree n:**

- (1) Let $z = f(x, y)$. If $f(tx, ty) = t^n f(x, y)$, then $f(x, y)$ is called as homogeneous function of degree n .
 - (2) If $z = f(x, y)$ is a homogeneous function of degree n then $f(x, y) = x^n f\left(\frac{y}{x}\right)$ or $f(x, y) = y^n f\left(\frac{x}{y}\right)$.
 - (3) Degree of homogeneous function may be any real number.
- **Euler's Theorem:** If $z = f(x, y)$ be a homogeneous function of degree n , then
 - (1) $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = nz$.
 - (2) $x \frac{\partial^2 z}{\partial x^2} + y \frac{\partial^2 z}{\partial y \partial x} = (n-1) \frac{\partial z}{\partial x}$.
 - (3) $x \frac{\partial^2 z}{\partial x \partial y} + y \frac{\partial^2 z}{\partial y^2} = (n-1) \frac{\partial z}{\partial y}$.
 - (4) $x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} = n(n-1)z$.
 - If $f(u)$ is homogeneous function in x and y of degree n then:
 - (1) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n \frac{f(u)}{f'(u)}$.
 - (2) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \alpha(u)[\alpha'(u) - 1]$ where $\alpha(u) = x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$.
 - If $v = f(u)$ where u is homogeneous function in x and y of degree n then:

$$x^2 \frac{\partial^2 v}{\partial x^2} + 2xy \frac{\partial^2 v}{\partial x \partial y} + y^2 \frac{\partial^2 v}{\partial y^2} = n(n-1)u\alpha'(u) + n^2 u^2 \alpha''(u) \text{ where } \alpha(u) = x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}.$$
 - If $f(x, y, z)$ is homogeneous function in x, y and z of degree n then:
 - (1) $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + z \frac{\partial f}{\partial z} = nf$.
 - (2) $x^2 \frac{\partial^2 f}{\partial x^2} + y^2 \frac{\partial^2 f}{\partial y^2} + z^2 \frac{\partial^2 f}{\partial z^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + 2yz \frac{\partial^2 f}{\partial y \partial z} + 2zx \frac{\partial^2 f}{\partial z \partial x} = n(n-1)f$.
 - If $u = f(r)$ and $r^2 = x^2 + y^2$ then $u_{xx} + u_{yy} = \frac{1}{r} f'(r) + f''(r)$.
 - If $u = f(r)$ and $r^2 = x^2 + y^2 + z^2$ then $u_{xx} + u_{yy} + u_{zz} = \frac{2}{r} f'(r) + f''(r)$.
 - If $u = r^n$ and $r^2 = x^2 + y^2 + z^2$ then $u_{xx} + u_{yy} + u_{zz} = n(n+1)u^{\frac{n-2}{n}} = n(n+1)r^{n-2}$.
